PROJECT BRIEF / SEMICON

Semicon Sales Analysis

Problem Statement and Dashboard Requirements

Background

Semicon sells computer hardware products through a customer base that includes brick-and-mortar and e-commerce sellers. Its transaction records span multiple products, customers, markets and dates. This project combines Python data preparation with a Tableau data model and interactive dashboards to make sales performance easier to review.

The supplied dataset contains 50,000 transaction rows and four supporting tables for customers, dates, markets and products. Transactions cover October 2017 through June 2020. The Tableau model relates the transaction fact table to these four dimensions.

Leadership and sales teams need a reliable view of sales scale, market performance, customer concentration, product contribution and monthly trends. Those questions are difficult to answer consistently when source fields contain formatting issues, multiple currencies and transaction values that require validation.

Business Questions

- How much sales activity is represented by the data, and how do sales and quantities change by month and year?
- Which markets contribute the most sales value and units, and where is sales coverage limited?
- Which customers account for the largest share of sales, and how concentrated is the customer portfolio?
- Which products lead by sales value, and what products contribute most to performance?
- Can executives compare KPI cards and market, customer, product and time views using one consistent sales definition?

Current Reporting Challenges

- The five related tables must be cleaned and connected consistently before cross-market and time-based analysis can be trusted.
- Source text includes hidden whitespace and carriage returns, the customer-name field contains a typo, and some market zones are missing.
- Transactions include non-positive sales amounts and multiple currency codes. These records and currency conversions need documented rules.
- Business users need an executive overview as well as a separate view of monthly and year-level trends.
- The Tableau headline revenue measure multiplies converted sales amount by quantity, while several comparison charts sum converted sales amount. The intended business definition and calculation basis need to be reconciled.

Analyst Role

As the Python and Tableau analyst, prepare and validate the data, build a clear relational model, define the reporting measures, and present findings in dashboards that support executive review. Document assumptions that affect interpretation, including currency conversion, transaction exclusions, period coverage and calculation definitions.

Analysis and Deliverables

Help business users move from a headline result to the market, customer, product or period detail behind it.

Data Preparation and Model

- Clean whitespace and hidden line breaks across text fields. Correct the customer-name field label, standardize date and numeric types, and rebuild date attributes from the transaction calendar.
- Remove exact duplicates and exclude rows with invalid dates or non-positive quantities or sales amounts. Retain traceability to original currency and source amount where converted amounts are used.
- Convert USD amounts using the stated project assumption of 96 INR per USD, and label that rate as a fixed analytical assumption rather than a historical exchange-rate series.
- Fill missing market zones with the project value "International"; validate dimension keys and transaction foreign keys before reporting.
- Relate the cleaned transaction table to customer, date, market and named-product dimensions. Use one documented sales-value definition consistently in all views.

Dashboard 1: Executive Overview

- Show headline sales value, units sold, active markets and active customers, with year and month filters.
- Compare sales value by market and units sold by market to surface market leaders and low-activity locations.
- Rank the top five customers and top five products by sales value.
- Provide clear titles, INR units, ranking context and definitions for active market and active customer counts.

Dashboard 2: Yearly Growth

- Show monthly sales value across the available transaction period, grouped by calendar year.
- Make month-to-month peaks, dips and recoveries visible and support filtering by year and month.
- Distinguish complete calendar years from partial periods. The records begin in October 2017 and end in June 2020, so 2017 and 2020 are partial.

Decision Support

The dashboards should allow leadership to identify where sales are concentrated, which accounts warrant focused attention, which products lead, and when performance changes. They should support investigation and planning; the data alone does not establish why a market, customer or product performed as it did.

SUCCESS CRITERIA / INTERPRETATION

Deliverables and Acceptance Criteria

Expected Deliverables

- A Python cleaning notebook and documented cleaned data outputs for transactions and the customer, date, market and product dimensions.
- A Tableau data model with the transaction table linked to the four corresponding dimensions.
- An interactive Tableau workbook containing an Executive Overview and a Yearly Growth dashboard, with navigation and shared date filters.
- An executive summary that explains dashboard findings, metric definitions, caveats and practical follow-up actions.

Acceptance Criteria

Area	Acceptance test
Data quality	Clean tables have consistent field types, unique dimension keys, no unresolved transaction-to-dimension key gaps, and documented exclusions.
Currency	Converted values use the documented fixed rate and display INR consistently. Original values and currency remain interpretable.
Measure consistency	The same sales-value definition and unit reconcile across KPI cards, market/customer/product rankings and time trends.
Dashboard clarity	Titles, filters, labels and footnotes make each metric, population, period and unit clear.
Time context	Partial years are identified and not presented as directly comparable to full years without qualification.
Decision usefulness	The dashboards answer the stated market, customer, product and trend questions and support drill-down through shared filters.

Scope note

This brief is reverse-engineered from the supplied notebook, source data, Tableau workbook, model image and dashboard screenshots. It describes the project's evidenced scope. The supplied materials do not include company strategy, cost, margin, inventory or customer profitability data.